

RAGHUL G

ASIC RTL / FPGA ENGINEER

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Experience

RTL Design Engineer I, Mirafra Software Technologies Pvt. Ltd | Feb 2025 - Present (Deployed in Qualcomm for Client Assignment)

Scope includes FPGA (Boolean Board, Black Board) and ASIC design. Worked across microarchitecture, RTL design, integration, low power design, linting, CDC, synthesis, STA, and FPGA flow. Involved in RISC-V based processor design using tools such as Spyglass (Lint & CDC), Design Compiler, Xilinx Vivado, ISE Design Suite, PrimeTime, and VCLP. Worked with **Verilog**, **C**, and **TCL**, with exposure to protocols like **DDR**, **AMBA - APB**, **AXI**.

Intern, FLSmith, On-Site | Aug 2023 – Sep 2023

Developed expertise in Internet of Things (IoT) with focus on edge and cloud computing. Learned and applied different models of the Software Development Life Cycle and IT terminologies. Implemented machine learning techniques for predictive analysis and gained hands-on experience in robotic process automation.

Projects

CPU Lite

Designed a basic RISC CPU (150 MHz) with data/program memory at 70 MHz. Developed microarchitecture, RTL, and integrated modules at top level. Implemented timing and control logic, verified with testbench, and resolved lint and CDC violations. Incorporated a 1K×32 SRAM buffer between CPU and data memory to efficiently handle cache hit/miss conditions for faster load and store operations.

Four Directional Traffic Light Controller

Designed an FSM-based controller with 5s green and 1s yellow lights. Implemented RTL, simulated, synthesized, and verified the design on Spartan-6 FPGA.

ALU

Developed a reconfigurable ALU supporting 13 arithmetic and 14 logical operations. Achieved single-cycle execution for most operations and 2-cycle for multiplication. Designed modular RTL with status flags for debugging and verified with testbench simulations.

AXI4-Lite based Master-Slave communication system

Developed an AXI4-Lite Master-Slave system in Verilog for LED and counter control using FSM-based handshaking. Verified through Vivado simulation and FPGA prototyping, showcasing efficient data transfer and in-depth understanding of AXI protocol operations.

MIPS ISA Simulator in C

Implemented a 32-bit simulator supporting R/I/J formats, load/store, and mul/div instructions. Designed register file, memory access, and error handling modules. And validated with 50+ test cases.

Skills

Languages: Verilog, C, Python, TCL

Design Competencies: Lint, CDC, Synthesis, FPGA, STA, Low Power Designs

Tools

Spyglass (Lint & CDC), Xilinx Vivado, Design Compiler, PrimeTime, VCLP

Education

B.Tech, Electronics and Communication Engineering, 2025, Vellore Institute of Technology, Chennai | CGPA: 8.44/10

XII, Science (PCMB), 2021, Velammal Vidhyashram | Percentage: 86.8%

X, General, 2019, Velammal Vidhyashram | Percentage: 89.8%

Professional Training

- Python Programming (IIT Bombay, Online)
- C Programming (IIT Bombay, Online)
- Artificial Intelligence & Machine Learning (Google Developers, Online)